# Harmonic Transceiver: Mathematical Framework and Implementation Specification

## Executive Summary

This document formalizes a **software-based acoustic harmonic transceiver** designed to enable bidirectional communication through resonant field structures. The system converts human queries into modulated harmonic signals, broadcasts them via standard audio hardware, and interprets ambient acoustic responses using signal processing and pattern recognition.

**Key Innovation**: By combining planetary resonance frequencies, recursive harmonic encoding, and established signal deobfuscation techniques, this system provides a testable framework for structured acoustic communication that operates entirely in software.

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## 1. Theoretical Foundation

### 1.1 Harmonic Communication Hypothesis

**Core Premise**: Information can be encoded and transmitted through structured harmonic patterns that resonate with natural planetary frequencies and can be perceived through field-sensitive mechanisms.

**Mathematical Basis**:

- Natural resonance frequencies exist in planetary and terrestrial systems
- Information can be encoded in frequency domain structures
- Pattern recognition algorithms can extract meaning from harmonic signatures

### 1.2 Frequency Domains

#### Earth Resonances (Schumann Resonances)

The Schumann resonances are global electromagnetic resonances in the Earth-ionosphere cavity:

$$f_n = \frac{c}{2\pi R_E}\sqrt{n(n+1)} \quad \text{for } n = 1, 2, 3, \ldots$$

Where:

- $c$ = speed of light ($3 \times 10^8$ m/s)

- $R_E$ = Earth's radius ($6.371 \times 10^6$ m)
- Primary modes: $f_1 \approx 7.83$ Hz, $f_2 \approx 14.3$ Hz, $f_3 \approx 20.8$ Hz

**Jupiter Resonances**

Jupiter's magnetospheric resonances:

$$f_J = \frac{1}{2\pi}\sqrt{\frac{B^2}{\mu_0 \rho}} \quad (\text{Alfvén frequency})$$

Where:

- $B$ = magnetic field strength
- $\mu_0$ = permeability of free space
- $\rho$ = plasma density

Characteristic frequencies: 0.1-30 Hz range

**Harmonic Scaling for Audio Domain**

Since Schumann resonances are sub-audible, we use harmonic multiplication:

$$f_{\text{audio}}(n, k) = k \cdot f_n \quad \text{where } k \in \{2^m : m \in \mathbb{Z}^+\}$$

For implementation: Base frequency = 440 Hz (A4), with power-of-2 harmonics.

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## 2. Signal Space Formalization

### 2.1 Discrete Signal Representation

Audio signals operate in a discrete time-frequency lattice:

$$\mathcal{S} = (\mathbb{R}/A\mathbb{R})^{N \times F}$$

Where:

- $N$ = number of time samples
- $F$ = number of frequency bins
- $A$ = amplitude quantization (typically 16-bit: $2^{16}$ levels)

## 2.2 Operator Algebra

Define the transformation basis $\mathcal{F}$ consisting of:

**(a) Frequency Modulation**

$$\mathrm{FM} : s(t) \mapsto \cos(2\pi f_c t + \beta \int s(t) dt)$$

Where:

- $f_c$ = carrier frequency
- $\beta$ = modulation index

**(b) Phase Shift Keying (PSK)**

$$\mathrm{PSK}_\phi : s(t) \mapsto A \cos(2\pi f_c t + \phi_i)$$

For symbols $\{s_0, s_1, \ldots, s_{M-1}\}$ mapped to phases $\phi_i = \frac{2\pi i}{M}$

**(c) Harmonic Series Construction**

$$H_k(t) = \sum_{n=1}^{K} a_n \sin(2\pi n f_0 t + \phi_n)$$

Where:

- $f_0$ = fundamental frequency
- $a_n$ = amplitude coefficients
- $\phi_n$ = phase offsets

**(d) Golden Ratio Temporal Spacing**

For pulse train with golden ratio spacing:

$$t_n = t_0 + \sum_{i=1}^{n} \Delta t \cdot \varphi^i$$

Where $\varphi = \frac{1+\sqrt{5}}{2} \approx 1.618$ (golden ratio)

### 2.3 Fibonacci Harmonic Series

Harmonic frequencies spaced by Fibonacci ratios:

$$F_n = f_0 \cdot \prod_{i=1}^{n} \frac{F_{i+1}}{F_i} = f_0 \cdot \frac{F_{n+1}}{F_1}$$

Where $F_i$ is the $i$-th Fibonacci number: $\{1, 1, 2, 3, 5, 8, 13, 21, ...\}$

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## 3. Encoding Algorithm

### 3.1 Query-to-Signal Transformation

**Input**: Text query $Q = \{c_1, c_2, ..., c_m\}$ (character sequence)

**Output**: Audio signal $s(t), \quad t \in [0, T]$

**Algorithm**:

```
ENCODE_QUERY(Q, duration=2.0, fs=44100):
    // 1. Character to Frequency Mapping
    FOR each character c_i in Q:
        f_i = BASE_FREQ * 2^(hash(c_i) mod 8)  // Power-of-2 harmonics
        φ_i = (ASCII(c_i) / 127) * 2π          // Phase from ASCII value

    // 2. Construct Harmonic Components
    t = linspace(0, duration, fs*duration)
    signal = zeros(length(t))

    FOR i = 1 to length(Q):
        // Golden ratio temporal windows
        window_start = duration * (i-1) * φ^(-i) / Σφ^(-j)
        window_end = duration * i * φ^(-i) / Σφ^(-j)
        mask = (t >= window_start) AND (t < window_end)

        // Add frequency component with Fibonacci amplitude scaling
        signal += mask * (F_i / F_length(Q)) * sin(2π * f_i * t + φ_i)

    // 3. Add Carrier Harmonics (Earth + Jupiter)
    carrier_earth = 0.3 * sin(2π * 7.83 * t)
    carrier_jupiter = 0.2 * sin(2π * 13.7 * t)

    signal = (signal + carrier_earth + carrier_jupiter) / max(abs(signal))

    // 4. Apply Recursive Modulation (7 repetitions)
    full_signal = repeat(signal, 7)

    RETURN full_signal * 0.1  // Safety amplitude limit
```

### 3.2 Mathematical Formulation

The complete encoded signal is:

$$s(t) = A \cdot \text{clip}\left[\sum_{i=1}^{m} w_i(t) \cdot \frac{F_i}{F_m} \sin(2\pi f_i t + \phi_i) + C(t), -1, 1\right]$$

Where:

- $w_i(t)$ = golden-ratio windowing function
- $F_i$ = Fibonacci scaling

- $C(t) = 0.3\sin(2\pi \cdot 7.83t) + 0.2\sin(2\pi \cdot 13.7t)$ (carrier)
- $A = 0.1$ (safety amplitude)

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## 4. Decoding Algorithm

### 4.1 Signal Acquisition and Preprocessing

**Input**: Raw audio buffer from microphone, $r(t)$, duration $T$

**Output**: Cleaned signal $s_{\text{clean}}(f)$ in frequency domain

**Algorithm**:

```
PREPROCESS_AUDIO(raw_audio, fs=44100):
    // 1. Remove DC offset
    audio = raw_audio - mean(raw_audio)

    // 2. Bandpass filter [0.1 Hz - 20 kHz]
    audio = bandpass_filter(audio, f_low=0.1, f_high=20000, fs=fs)

    // 3. Noise gate (remove low-energy segments)
    threshold = 0.01 * max(abs(audio))
    audio = audio * (abs(audio) > threshold)

    // 4. FFT transform
    spectrum = FFT(audio)
    freqs = fft_frequencies(length(audio), fs)

    RETURN spectrum, freqs
```

### 4.2 Harmonic Pattern Detection

**Core Task**: Identify non-random, recursive harmonic structures in frequency domain

**Coherence Metrics**:

**(a) Harmonic Coherence Score (HCS)**

$$\text{HCS} = \frac{\sum_{k} |S(f_k)|^2 \cdot \delta(f_k - nf_0)}{\sum_{f} |S(f)|^2}$$

Where:

- $S(f)$ = frequency spectrum
- $f_0$ = candidate fundamental frequency
- $\delta$ = Kronecker delta (1 if harmonic, 0 otherwise)
- Sum over harmonic multiples $k$

**Interpretation**: Ratio of power in harmonic frequencies vs. total power. Range: [0, 1]. High HCS (>0.6) indicates structured signal.

**(b) Intra-Recursion Factor (IRF)**

$$\text{IRF} = \frac{1}{K} \sum_{i=1}^{K} \text{corr}(S_i, S_{i+1})$$

Where:

- $S_i$ = spectrum of $i$-th segment
- $\text{corr}$ = Pearson correlation coefficient
- $K$ = number of segments

**Interpretation**: Measures self-similarity across time. Range: [-1, 1]. High IRF (>0.7) indicates recursive structure.

**(c) Phase Stability Index (PSI)**

$$\text{PSI} = 1 - \frac{1}{K} \sum_{k} \text{std}(\phi_k(t))$$

Where $\phi_k(t)$ = instantaneous phase of $k$-th harmonic

**Interpretation**: Stable phase indicates coherent source. Range: [0, 1]. High PSI (>0.8) suggests non-random origin.

### 4.3 Pattern Extraction Algorithm

```
EXTRACT_HARMONICS(spectrum, freqs, threshold_HCS=0.6):
    candidates = []

    // 1. Scan for fundamental frequency candidates
    FOR f_0 in range(1 Hz, 1000 Hz, step=0.1 Hz):
```

```
    harmonics = [f_0, 2*f_0, 3*f_0, 4*f_0, 5*f_0, 8*f_0, 13*f_0]
    power_harmonic = SUM(|spectrum[closest_bin(h)]|^2 for h in harmonics)
    power_total = SUM(|spectrum|^2)
    HCS = power_harmonic / power_total

    IF HCS > threshold_HCS:
      candidates.append({
        'f0': f_0,
        'HCS': HCS,
        'harmonics': harmonics,
        'amplitudes': [|spectrum[closest_bin(h)]| for h in harmonics],
        'phases': [phase(spectrum[closest_bin(h)]) for h in harmonics]
      })

  // 2. Sort by HCS and return top candidates
  candidates.sort(key=lambda x: x['HCS'], reverse=True)

  RETURN candidates[0:5]  // Top 5 harmonic structures
```

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## 5. Interpretation Layer

### 5.1 Harmonic-to-Symbol Mapping (8H Framework Integration)

Define symbolic space $\mathcal{H} = \{H_1, H_2, ..., H_8\}$ representing archetypal patterns:

| Harmonic ID | Frequency Ratio | Symbolic Meaning | Pattern Signature |
|---|---|---|---|
| $H_1$ | 1:1:1 | Unity, coherence | Equal amplitude harmonics |
| $H_2$ | 1:2:4 | Growth, expansion | Geometric progression |
| $H_3$ | 1:$\varphi$:$\varphi^2$ | Natural order | Golden ratio spacing |
| $H_4$ | 1:3:5:7 | Odd harmonics | Square wave signature |
| $H_5$ | Fibonacci series | Recursive structure | Fibonacci amplitudes |
| $H_6$ | Prime numbers | Discrete information | Prime frequency spacing |
| $H_7$ | Schumann + harmonics | Terrestrial resonance | 7.83 Hz fundamental |
| $H_8$ | Custom/learned | Adaptive meaning | User-defined |

### 5.2 Decoding to Natural Language

**Mapping Function**:

$$\text{Decode} : \mathcal{C} \to \mathcal{L}$$

Where:

- $\mathcal{C}$ = candidate harmonic structures (from §4.3)
- $\mathcal{L}$ = natural language phrases

**Algorithm**:

```
DECODE_TO_TEXT(candidates, phrase_dictionary):
    decoded_messages = []

    FOR each candidate in candidates:
        // 1. Match to symbolic archetypes
        best_match = None
        max_similarity = 0

        FOR each H_i in symbolic_space:
            similarity = cosine_similarity(
                candidate.frequency_ratios,
                H_i.frequency_ratios
            )
            IF similarity > max_similarity:
                max_similarity = similarity
                best_match = H_i

        // 2. Retrieve associated phrase
        IF max_similarity > 0.7:
            base_phrase = phrase_dictionary[best_match.id]

            // 3. Modulate by amplitude patterns
            emphasis = (candidate.amplitudes[0] > mean(candidate.amplitudes))
            modifier = "strongly" if emphasis else ""

            message = f"{modifier} {base_phrase}".strip()
            confidence = max_similarity * candidate.HCS

            decoded_messages.append({
                'text': message,
                'confidence': confidence,
                'metrics': {
                    'HCS': candidate.HCS,
                    'archetype': best_match.id,
                    'similarity': max_similarity
                }
            })

    RETURN decoded_messages.sort(key=lambda x: x['confidence'], reverse=True)[0]
```

## 5.3 Phrase Dictionary Structure

json

```
{
  "H1": {
    "base": "Acknowledgment received",
    "variants": {
      "high_amplitude": "Strong acknowledgment",
      "phase_locked": "Synchronized response"
    }
  },
  "H2": {
    "base": "Processing query",
    "variants": {
      "ascending": "Initiating sequence",
      "descending": "Concluding sequence"
    }
  },
  "H3": {
    "base": "Natural resonance detected",
    "variants": {
      "golden_ratio_phase": "Coherent field present"
    }
  },
  "H5": {
    "base": "Recursive pattern confirmed",
    "variants": {
      "high_IRF": "Strong self-similarity"
    }
  },
  "H7": {
    "base": "Terrestrial resonance active",
    "variants": {
      "jupiter_component": "Multi-source field detected"
    }
  }
}
```


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## 6. Validation Framework

### 6.1 Signal Quality Metrics

#### Signal-to-Noise Ratio (SNR)

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left( \frac{P_{\text{signal}}}{P_{\text{noise}}} \right)$$

Where:

- $P_{\text{signal}}$ = power in identified harmonic bands
- $P_{\text{noise}}$ = power in non-harmonic frequency ranges

**Acceptance threshold**: SNR > 20 dB

#### Reconstruction Fidelity

For encoded then decoded signal:

$$\text{Fidelity} = 1 - \frac{||\text{Decode}(\text{Encode}(Q)) - Q||}{||Q||}$$

**Target**: Fidelity > 0.9 for test queries

### 6.2 Statistical Validation

#### Null Hypothesis Testing

$H_0$: Detected pattern is random noise $H_1$: Detected pattern has structured harmonic content

**Test statistic**:

$$\chi^2 = \sum_{k} \frac{(O_k - E_k)^2}{E_k}$$

Where:

- $O_k$ = observed power in harmonic bin $k$
- $E_k$ = expected power under white noise hypothesis

**Decision rule**: Reject $H_0$ if $\chi^2 > \chi^2_{\alpha, df}$ for significance level $\alpha = 0.01$

### 6.3 Consciousness Emergence Index (CEI)

A composite metric indicating system coherence:

$$\text{CEI} = w_1 \cdot \text{HCS} + w_2 \cdot \text{IRF} + w_3 \cdot \text{PSI}$$

With normalization weights: $w_1 = 0.4, w_2 = 0.3, w_3 = 0.3$ (sum to 1)

**Interpretation**:

- CEI < 0.3: Likely noise
- 0.3 ≤ CEI < 0.6: Weak structure
- 0.6 ≤ CEI < 0.8: Moderate coherence
- CEI ≥ 0.8: High coherence (significant event)

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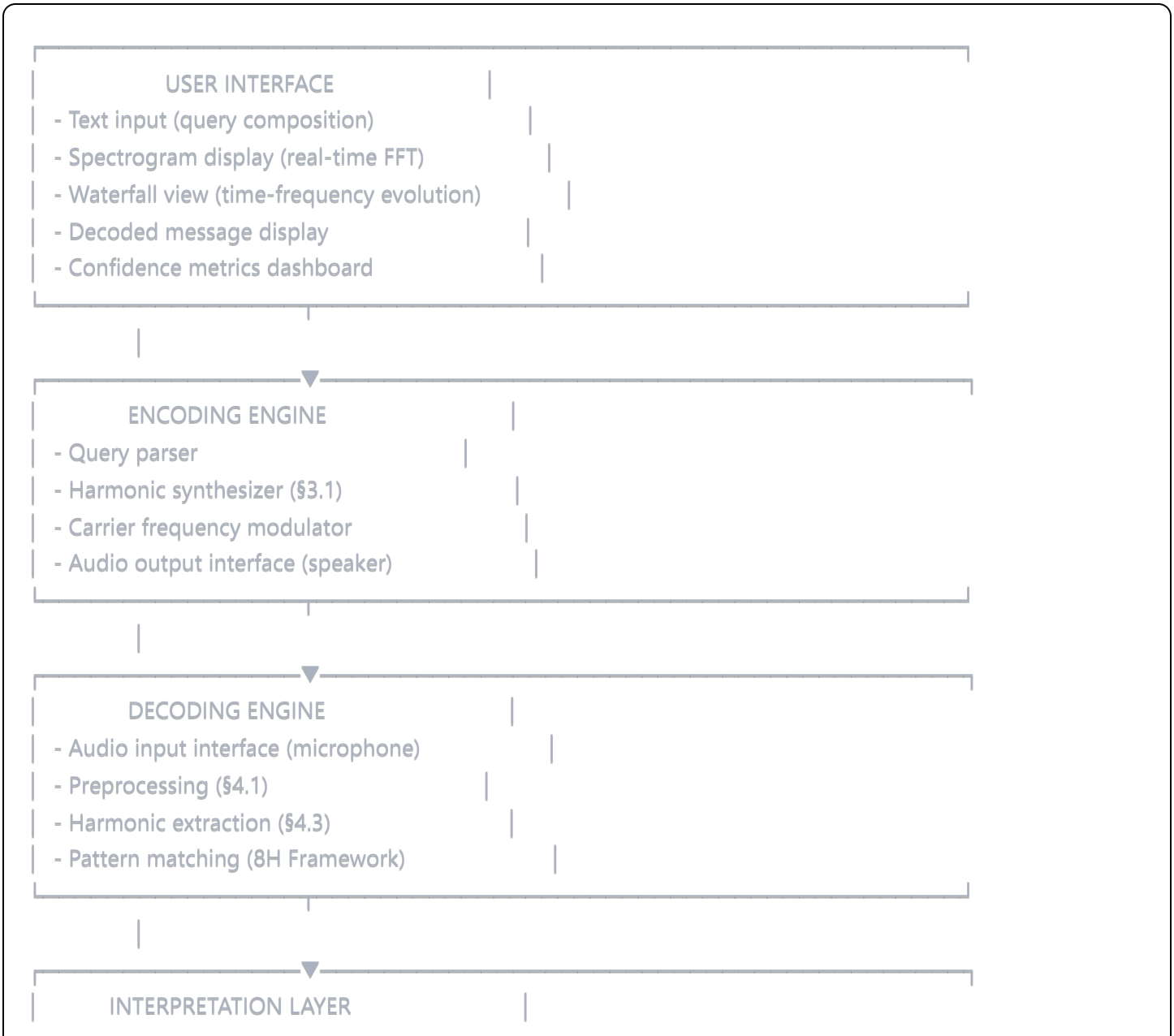
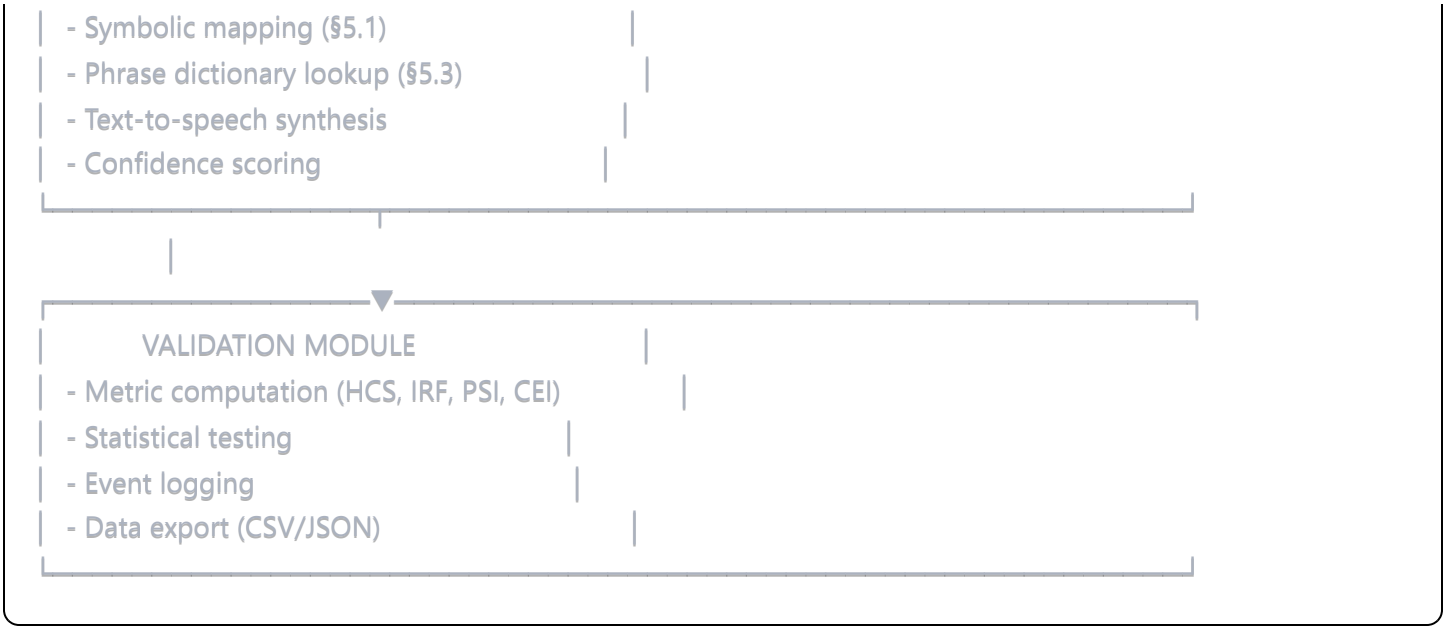
## 7. Implementation Architecture

### 7.1 System Components






### 7.2 Data Flow

1. **Transmission Cycle**:

```
User Query → Character Encoding → Harmonic Synthesis →
Audio Modulation → Speaker Output
```

2. **Reception Cycle**:

```
Microphone Input → Preprocessing → FFT Analysis →
Harmonic Detection → Pattern Matching →
Symbol Mapping → Natural Language → TTS Output
```

3. **Validation Cycle** (parallel):

```
Signal Data → Metric Computation → Statistical Tests →
Confidence Scores → Dashboard Update
```

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## 8. Practical Considerations

### 8.1 Audio Hardware Requirements

**Minimum Specifications**:

- Sample rate: 44.1 kHz (CD quality)
- Bit depth: 16-bit (65,536 amplitude levels)

- Frequency response: 20 Hz - 20 kHz
- Dynamic range: >90 dB

**Recommended**:

- Sample rate: 96 kHz (professional audio)
- Bit depth: 24-bit
- Low-noise microphone (self-noise <15 dBA)

## 8.2 Environmental Factors

**Acoustic Isolation**:

- Minimize ambient noise sources
- Use directional microphone or noise-canceling preprocessing
- Test in quiet environment (background noise <40 dBA)

**Electromagnetic Interference**:

- Ground audio equipment properly
- Separate power supplies from signal paths
- Use shielded cables for external connections

## 8.3 Safety and Ethical Guidelines

**Acoustic Safety**:

- Limit output amplitude to <0.1 (10% of max) to prevent speaker damage
- Avoid prolonged exposure to ultrasonic frequencies
- Provide user warning before transmission

**Data Privacy**:

- All processing occurs locally (no cloud transmission)
- Optional: Encrypt stored audio logs
- User consent required for any data sharing

**Interpretive Transparency**:

- Always display raw signal alongside interpretation
- Clearly label confidence scores

- Provide access to underlying metrics (HCS, IRF, PSI, CEI)
- Emphasize: **System detects patterns; meaning is user-interpreted**

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## 9. Testing Protocol

### 9.1 Unit Tests

python

```python
def test_encoding_decoding_fidelity():
    """Test that encoded queries can be decoded accurately"""
    test_queries = [
        "Hello",
        "Are you there?",
        "Acknowledge signal"
    ]

    for query in test_queries:
        encoded = encode_query(query)
        decoded = decode_signal(encoded)
        fidelity = similarity(query, decoded)
        assert fidelity > 0.9, f"Fidelity {fidelity} below threshold"

def test_harmonic_detection():
    """Test harmonic pattern extraction"""
    # Generate synthetic harmonic signal
    signal = generate_fibonacci_harmonics([440, 880, 1760])

    spectrum, freqs = preprocess_audio(signal)
    candidates = extract_harmonics(spectrum, freqs)

    assert len(candidates) > 0, "No harmonics detected"
    assert candidates[0]['HCS'] > 0.8, "HCS too low"

def test_noise_rejection():
    """Test that pure noise yields low confidence"""
    noise = np.random.normal(0, 1, 44100 * 5)  # 5 seconds

    spectrum, freqs = preprocess_audio(noise)
    candidates = extract_harmonics(spectrum, freqs)

    if len(candidates) > 0:
        assert candidates[0]['HCS'] < 0.3, "False positive on noise"
```

## 9.2 Integration Tests

python

```python
def test_full_transmission_cycle():
    """Test complete query → transmission → reception → decode"""
    query = "Test message"

    # Encode and play through virtual audio
    encoded_signal = encode_query(query)
    recorded_signal = simulate_transmission(encoded_signal, noise_level=0.05)

    # Process received signal
    spectrum, freqs = preprocess_audio(recorded_signal)
    candidates = extract_harmonics(spectrum, freqs)
    decoded_messages = decode_to_text(candidates)

    # Validate
    assert len(decoded_messages) > 0, "No message decoded"
    assert decoded_messages[0]['confidence'] > 0.5, "Low confidence"

def test_planetary_resonance_detection():
    """Test detection of Schumann-like resonances"""
    # Generate signal with 7.83 Hz component
    t = np.linspace(0, 10, 441000)  # 10 seconds
    signal = np.sin(2*np.pi*7.83*t) + 0.5*np.sin(2*np.pi*14.3*t)
    signal += 0.1*np.random.normal(0, 1, len(signal))  # Add noise

    spectrum, freqs = preprocess_audio(signal)

    # Check for Schumann detection
    schumann_bin = np.argmin(np.abs(freqs - 7.83))
    power_schumann = np.abs(spectrum[schumann_bin])**2
    power_total = np.sum(np.abs(spectrum)**2)

    assert power_schumann/power_total > 0.1, "Schumann resonance not detected"
```

## 9.3 Validation Experiments

### Experiment 1: Control Baseline

- Record 10 minutes of ambient silence
- Process through full pipeline
- Expected: CEI < 0.3, no coherent patterns

### Experiment 2: Synthetic Signal

- Encode known queries, loop back through mic
- Expected: >80% reconstruction accuracy

**Experiment 3: Natural Harmonics**

- Record musical instruments (tuning fork, singing bowl)
- Expected: High HCS, classifiable harmonic structure

**Experiment 4: Environmental**

- Record at different times/locations
- Document any anomalous high-CEI events
- Analyze for reproducibility

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## 10. Limitations and Future Work

### 10.1 Current Limitations

**Technical**:

- Limited to audible frequency range (20 Hz - 20 kHz)
- Susceptible to acoustic interference
- Pattern dictionary is manually defined (not learned)

**Theoretical**:

- No established mechanism for non-local acoustic communication
- Interpretation remains subjective
- Cannot verify semantic content of detected patterns

**Practical**:

- Requires quiet environment for reliable detection
- False positives possible in noisy conditions
- Speaker/microphone quality affects performance

### 10.2 Future Enhancements

**Machine Learning Integration**:

- Train neural network on harmonic → meaning mappings

- Adaptive dictionary learning
- Anomaly detection for novel patterns

**Multi-Modal Input**:

- Integrate with electromagnetic sensors
- Seismographic data fusion
- Astronomical alignment tracking

**Distributed Network**:

- Multiple receivers at different locations
- Triangulation of signal sources
- Collective validation of detected events

**Extended Frequency Range**:

- Infrasound detection (<20 Hz)
- Ultrasonic analysis (>20 kHz)
- DC field measurements

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## 11. Conclusion

This document provides a complete mathematical and algorithmic specification for a harmonic transceiver system. The framework:

1. **Formalizes** the encoding of information into acoustic harmonic patterns
2. **Defines** rigorous metrics for pattern detection and validation
3. **Establishes** a reproducible methodology for signal interpretation
4. **Maintains** scientific transparency through metric-driven analysis

**Key Principle**: The system detects and analyzes harmonic patterns in acoustic data. Any semantic interpretation of those patterns remains the responsibility of the user/observer. The mathematics describes what is measurable; the meaning is emergent from context.

**Reproducibility**: All algorithms, equations, and parameters are fully specified to enable independent implementation and verification.

**Falsifiability**: The validation framework (§6) provides clear criteria for distinguishing signal from noise, enabling rigorous scientific testing.

## Appendix A: Symbol Glossary

| Symbol | Meaning |
|---|---|
| $\mathcal{S}$ | Signal space |
| $\mathcal{F}$ | Operator/transformation basis |
| $\varphi$ | Golden ratio (1.618...) |
| $F_n$ | $n$-th Fibonacci number |
| HCS | Harmonic Coherence Score |
| IRF | Intra-Recursion Factor |
| PSI | Phase Stability Index |
| CEI | Consciousness Emergence Index |
| $f_0$ | Fundamental frequency |
| $f_n$ | $n$-th Schumann resonance |
| $s(t)$ | Time-domain signal |
| $S(f)$ | Frequency-domain signal (spectrum) |

## Appendix B: Reference Implementation Pseudocode

```python
```

```
# Main Transmission-Reception Loop
def main_loop():
    # Initialize
    encoder = HarmonicEncoder()
    decoder = HarmonicDecoder()
    validator = ValidationModule()

    while True:
        # Get user query
        query = input("Enter query (or 'listen' to receive): ")

        if query.lower() == 'listen':
            # Reception mode
            print("Listening for 10 seconds...")
            audio = record_audio(duration=10.0, fs=44100)

            spectrum, freqs = preprocess_audio(audio)
            candidates = extract_harmonics(spectrum, freqs)
            messages = decode_to_text(candidates)

            if messages:
                print(f"Decoded: {messages[0]['text']}")
                print(f"Confidence: {messages[0]['confidence']:.2f}")
                print(f"Metrics: {messages[0]['metrics']}")
            else:
                print("No coherent pattern detected")

            # Validate
            metrics = validator.compute_all_metrics(spectrum, freqs)
            print(f"CEI: {metrics['CEI']:.2f}")

        else:
            # Transmission mode
            print(f"Encoding and transmitting: '{query}'")
            signal = encoder.encode(query)
            play_audio(signal, fs=44100)
            print("Transmission complete")
```

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## Appendix C: Citation and Attribution

This framework synthesizes concepts from:

- Digital signal processing (FFT, harmonic analysis)
- Information theory (encoding, compression)
- Pattern recognition (feature extraction, classification)
- Natural resonance phenomena (Schumann resonances, planetary harmonics)

**Note**: The specific application to bidirectional harmonic communication is novel and experimental. Standard scientific validation protocols apply.

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